

Bluetooth:-

Bluetooth is also known as IEEE 802.15 standard & specification that uses low power radio communications to link phones, computers and other network devices over a short distance without using any type of connecting wires.

- * It is used to send & receive data to connected devices present across a certain distance using a band of 2.4 to 2.485 GHz.
- * In bluetooth technology, the wireless signals transmit data and flies over a short distance, typically up to 30 feet or 10 meters.
- * It provides data rates up to 1 Mbps & 3 Mbps depending upon the version.

Bluetooth Architecture:

- * In bluetooth technology, the network of bluetooth consists of a Personal Area Network.
- * A bluetooth's architecture is also called a "Piconet" because it is made of multiple networks.
- * It contains a minimum of 2 to a maximum of 8 bluetooth peer devices.
- * It usually contains a single master and up to 7 slaves.
- Piconet provides the technology which facilitates data transmission based on its nodes, i.e., Master node and Slave nodes.

Applications:-

- * It can be used in laptops, and in wireless PC, printer.
- * It can be used in wireless headsets, wireless PANs & LANs
- * It can connect a digital camera wirelessly to a mobile phone.
- * It can transfer data in terms of videos, songs, photographs, or files from one cellphone to another cellphone or computer
- * It is used in the sectors of Medical health care, sports and fitness, Military.

Bluetooth Frame structure:-

The various fields of bluetooth frame format are:

72 bits	54 bits	0-2744 bits
Access Code	Header	Data

3 Times $\times 18 = 54$ bits

Address	Type	F	A	S	Checksum
3 bits	4 bits	1 bit	1 bit	1 bit	8 bits.

Optimized Link State Routing (OLSR) :-

- Avoid ~~unfess~~ unnecessary transmission of LS Packet
- Each node decide which of its neighbors can flood LS packet. These nodes known as MPR (Multi Point Relay)
- Only MPR can retransmit, other node don't transmit

OLSR Terminology:-

1. Neighbor Node:

Node A is neighbor node of B, if can hear A

2. 2 Hop Neighbor:

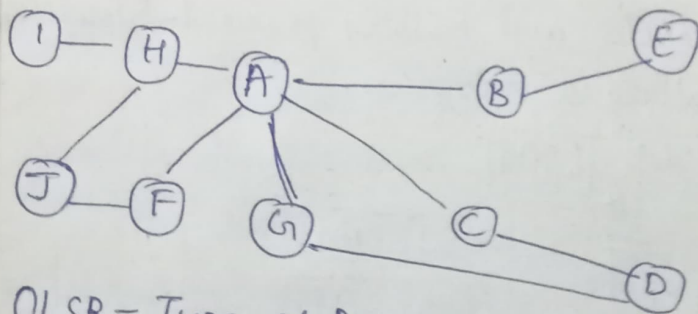
Node which can be heard by neighbor.

3. Strict 2 Hop Neighbor:

Node which can be heard by neighbor excluding node itself and its neighbor.

4. MultiPoint Relay:-

Node which is selected by its 1 hop neighbor to retransmit all broadcast message received by it



OLSR - Type Of Packets:-

1. Hello packets
2. Topology Control Packets
3. MLD Packets.